

DEUTSCHE BANK INDIA

Retail Banking — Digital Payments

API INTEGRATION SPECIFICATION

UPI Enablement Platform

Endpoint-Level Contracts Across CBS, CIF, PSP Switch, NPCI, and Supporting Systems

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Document Control

1.1 Version History

Version	Date	Author	Description
0.1	10-Jul-2026	Business Analyst	Initial endpoint inventory derived from FRS Section 12
1.0	date TBD	Business Analyst	Baseline for Integration Team technical design

1.2 Relationship to Other Deliverables

This specification elaborates the Integration Architecture and API-level requirements (API-REQ-01 to API-REQ-09) defined in Section 12 of the Functional Requirements Specification (FRS v1.0) into endpoint-level contracts: methods, paths, request/response fields, and error codes. It is a business-analyst-authored contract intended to accelerate technical design; the Integration Team and API Developers own the final OpenAPI/Swagger specification and any protocol-level detail (e.g., exact NPCI XML/ISO 8583 message structures) not covered here.

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2. API Conventions

These conventions apply to all internal APIs listed in this document, consistent with FRS Section 12.3 (API-REQ-01 to API-REQ-09). NPCI-facing message formats follow the NPCI UPI API specification and are documented here only at a business/field level, not at the wire-protocol level.

2.1 Base URL Structure

All internal APIs are versioned and namespaced by owning system: `https://internal-api.dbindia.co.in/{system}/v1/{resource}`. All requests and responses use JSON, except NPCI-bound messages which follow NPCI's mandated XML/ISO 20022-derived format via the PSP Switch adapter layer.

2.2 Authentication

Aspect	Convention
Internal service-to-service	Mutual TLS (mTLS) or OAuth 2.0 client-credentials grant, per API-REQ-03
Mobile App → API Gateway	Bearer token (session JWT) issued by the Authentication Service at login
PSP Switch → NPCI	NPCI-issued certificates and encryption keys per NPCI PSP onboarding requirements

2.3 Common Request Headers

Header	Description
X-Correlation-Id	Unique end-to-end transaction/request identifier, propagated across all systems (API-REQ-04)
Authorization	Bearer token or mTLS client certificate reference
X-Customer-Id	Internal customer identifier (where applicable, post-authentication)
Content-Type	application/json (unless otherwise specified)
Accept-Version	API version requested, e.g., v1

2.4 Common Response Envelope

Field	Type	Description
status	String	SUCCESS FAILURE PENDING
data	Object	Endpoint-specific payload (present on SUCCESS)
error_code	String	Present on FAILURE; internal error code per FRS Section 13 catalogue
error_message	String	Human-readable error description (not necessarily customer-facing)
correlation_id	String	Echoes the X-Correlation-Id for traceability

2.5 Timeout and Retry Policy

Each endpoint below specifies a target timeout aligned to overall transaction SLA (FRS NFR-PERF-01). Idempotent endpoints (marked below) may be safely retried by the caller using the same X-Correlation-Id; non-idempotent

endpoints (e.g., debit/credit posting) must not be retried without first checking transaction status, per API-REQ-05 and API-REQ-09.

3. Authentication Service

Validates the customer's mobile banking session and supports biometric re-authentication for step-up scenarios (e.g., PIN reset, high-value transaction confirmation).

POST /auth/v1/session/validate

Attribute	Value		
Purpose	Validate an active mobile banking session token before allowing entry into the UPI module		
Owning System	Authentication Service		
Target Timeout	≤ 500 ms		
Idempotent	Yes		
Field	Type	Mandatory	Description
session_token	String	M	Request: JWT session token issued at mobile banking login
device_id	String	M	Request: Device identifier for binding cross-check
is_valid	Boolean	-	Response: Whether the session is currently valid
customer_id	String	-	Response: Resolved internal customer identifier
expires_at	DateTime	-	Response: Session expiry timestamp

POST /auth/v1/biometric/verify

Attribute	Value		
Purpose	Perform step-up biometric re-authentication for sensitive actions (PIN reset, large transactions)		
Owning System	Authentication Service		
Target Timeout	≤ 1.5 s		
Idempotent	Yes		
Field	Type	Mandatory	Description
customer_id	String	M	Request: Internal customer identifier
biometric_assertion	String	M	Request: Signed assertion from device biometric SDK
verified	Boolean	-	Response: Verification result

4. Customer Information File (CIF)

Provides customer profile and KYC status used during registration and periodic eligibility re-checks.

GET /cif/v1/customers/{customerId}

Attribute	Value
Purpose	Fetch customer profile attributes for registration and support screens
Owning System	CIF
Target Timeout	≤ 800 ms
Idempotent	Yes

Field	Type	Mandatory	Description
customerId	String (path)	M	Request: Internal customer identifier
full_name	String	-	Response: Customer's registered full name
kyc_status	Enum	-	Response: VERIFIED PENDING EXPIRED
registered_mobile_no	String	-	Response: Mobile number on record, used for OTP verification match
customer_segment	String	-	Response: Internal segmentation code (not customer-facing)

GET /cif/v1/customers/{customerId}/kyc-status

Attribute	Value
Purpose	Lightweight KYC status check used to gate UPI registration eligibility
Owning System	CIF
Target Timeout	≤ 500 ms
Idempotent	Yes

Field	Type	Mandatory	Description
customerId	String (path)	M	Request: Internal customer identifier
kyc_status	Enum	-	Response: VERIFIED PENDING EXPIRED
last_verified_date	Date	-	Response: Date of last KYC verification

5. Core Banking System (CBS)

Provides account discovery, balance enquiry, debit/credit posting, and debit card validation. This is the platform's most latency-sensitive integration and is central to meeting NPCI transaction SLAs (FRS NFR-PERF-01).

GET /cbs/v1/customers/{customerId}/accounts

Attribute	Value		
Purpose	List all accounts held by the customer, for eligibility filtering during registration and account linking		
Owning System	CBS		
Target Timeout	≤ 1.5 s		
Idempotent	Yes		
Field	Type	Mandatory	Description
customerId	String (path)	M	Request: Internal customer identifier
accounts[].account_number	String	-	Response: Masked account number
accounts[].account_type	Enum	-	Response: SAVINGS CURRENT LOAN FD CREDIT_CARD RD
accounts[].account_status	Enum	-	Response: ACTIVE DORMANT FROZEN CLOSED
accounts[].ifsc_code	String	-	Response: Branch IFSC code

GET /cbs/v1/accounts/{accountId}/balance

Attribute	Value		
Purpose	Real-time balance enquiry, PIN-authenticated at the mobile app layer before this call is made		
Owning System	CBS		
Target Timeout	≤ 2 s		
Idempotent	Yes		
Field	Type	Mandatory	Description
accountId	String (path)	M	Request: Internal account identifier
available_balance	Decimal	-	Response: Funds available for withdrawal/transfer
current_balance	Decimal	-	Response: Ledger balance including holds
as_of	DateTime	-	Response: Timestamp of balance snapshot

POST /cbs/v1/accounts/{accountId}/debit

Attribute	Value
Purpose	Post a debit against the customer's account as part of an outbound UPI transaction
Owning System	CBS
Target Timeout	≤ 2 s

Attribute	Value		
Idempotent	No — requires status check before retry		
Field	Type	Mandatory	Description
accountId	String (path)	M	Request: Internal account identifier
amount	Decimal	M	Request: Debit amount
transaction_ref	String	M	Request: Unique transaction reference (idempotency key)
narration	String	O	Request: Transaction note/remark
debit_status	Enum	-	Response: SUCCESS INSUFFICIENT_FUNDS ACCOUNT_BLOCKED FAILED
cbs_reference	String	-	Response: CBS-side posting reference for reconciliation

POST /cbs/v1/accounts/{accountId}/credit

Attribute	Value		
Purpose	Post a credit against the customer's account (incoming transaction or reversal)		
Owning System	CBS		
Target Timeout	≤ 2 s		
Idempotent	No — requires status check before retry		
Field	Type	Mandatory	Description
accountId	String (path)	M	Request: Internal account identifier
amount	Decimal	M	Request: Credit amount
transaction_ref	String	M	Request: Unique transaction reference (idempotency key)
credit_type	Enum	M	Request: INBOUND_PAYMENT REVERSAL REFUND
credit_status	Enum	-	Response: SUCCESS FAILED
cbs_reference	String	-	Response: CBS-side posting reference

POST /cbs/v1/cards/validate

Attribute	Value		
Purpose	Validate debit card details (last 6 digits + expiry) as part of UPI PIN setup/reset		
Owning System	CBS		
Target Timeout	≤ 1 s		
Idempotent	Yes		
Field	Type	Mandatory	Description
accountId	String	M	Request: Internal account identifier
card_last6	String	M	Request: Last 6 digits of debit card number

Attribute		Value	
card_expiry	String	M	Request: Card expiry, format MMY
card_valid	Boolean	-	Response: Whether the card details match and card is active

6. SMS Gateway

Delivers OTPs for mobile verification, PIN setup/reset, and serves as the fallback channel for security-critical notifications.

POST /sms/v1/otp/send

Attribute	Value		
Purpose	Dispatch a 6-digit OTP to the customer's registered mobile number		
Owning System	SMS Gateway		
Target Timeout	≤ 3 s (dispatch, not delivery)		
Idempotent	No — generates a new OTP each call		
Field	Type	Mandatory	Description
mobile_no	String	M	Request: Registered mobile number (masked in logs)
purpose	Enum	M	Request: REGISTRATION PIN_SETUP PIN_RESET
otp_reference	String	-	Response: Reference ID for the OTP session (used in verify call)
expires_in_seconds	Integer	-	Response: OTP validity window (300s standard)

POST /sms/v1/otp/verify

Attribute	Value		
Purpose	Validate a customer-entered OTP against the issued value		
Owning System	SMS Gateway		
Target Timeout	≤ 500 ms		
Idempotent	Yes		
Field	Type	Mandatory	Description
otp_reference	String	M	Request: Reference ID from the send call
otp_value	String	M	Request: Customer-entered 6-digit code
verified	Boolean	-	Response: Verification result
attempts_remaining	Integer	-	Response: Remaining attempts before cooldown (FR-REG-02 related)

7. Notification Service

Dispatches push and SMS notifications for all platform events per the Notification Event Catalogue in FRS Section 11.1, and manages customer notification preferences.

POST /notify/v1/send

Attribute	Value		
Purpose	Trigger a notification for a platform event, per the mandatory 5-second delivery target (BR-NOT-01)		
Owning System	Notification Service		
Target Timeout	≤ 5 s (dispatch)		
Idempotent	No — each call sends a new notification		
Field	Type	Mandatory	Description
customer_id	String	M	Request: Internal customer identifier
event_type	Enum	M	Request: e.g., PAYMENT_SUCCESS, COLLECT_RECEIVED, SECURITY_ALERT (full list in FRS 11.1)
channel	Enum	M	Request: PUSH SMS PUSH_AND_SMS
payload	Object	M	Request: Event-specific merge fields for the message template
delivery_status	Enum	-	Response: QUEUED DELIVERED FAILED

GET /notify/v1/{customerId}/preferences

Attribute	Value		
Purpose	Fetch the customer's current notification preferences		
Owning System	Notification Service		
Target Timeout	≤ 500 ms		
Idempotent	Yes		
Field	Type	Mandatory	Description
customerId	String (path)	M	Request: Internal customer identifier
marketing_opt_in	Boolean	-	Response: Whether marketing/cross-sell notifications are enabled

PUT /notify/v1/{customerId}/preferences

Attribute	Value		
Purpose	Update the customer's notification preferences (marketing/cross-sell only, per BR-NOT-02)		
Owning System	Notification Service		
Target Timeout	≤ 500 ms		

Attribute	Value		
Idempotent	Yes		
Field	Type	Mandatory	Description
customerId	String (path)	M	Request: Internal customer identifier
marketing_opt_in	Boolean	M	Request: New preference value
updated	Boolean	-	Response: Confirmation of update

8. PSP Switch

The PSP Switch is the single orchestration layer between the Bank's internal systems and the NPCI UPI network (FRS Section 12.2). It exposes the endpoints below to the Mobile App / API Gateway layer and internally translates them into NPCI-mandated message formats.

POST /psp/v1/profile/create

Attribute	Value		
Purpose	Create a UPI profile and assign a VPA during registration		
Owning System	PSP Switch		
Target Timeout	≤ 3 s		
Idempotent	Yes (via transaction_ref)		
Field	Type	Mandatory	Description
customer_id	String	M	Request: Internal customer identifier
mobile_no	String	M	Request: Verified registered mobile number
preferred_vpa_handle	String	O	Request: Customer-requested custom handle (FR-REG-10)
linked_accounts	Array	M	Request: List of account identifiers with default flag
vpa	String	-	Response: Assigned/confirmed VPA
profile_status	Enum	-	Response: ACTIVE FAILED

POST /psp/v1/vpa/resolve

Attribute	Value		
Purpose	Resolve a mobile number or VPA to recipient account holder details prior to payment confirmation		
Owning System	PSP Switch		
Target Timeout	≤ 2 s		
Idempotent	Yes		
Field	Type	Mandatory	Description
identifier_type	Enum	M	Request: MOBILE VPA
identifier_value	String	M	Request: The mobile number or VPA entered by the customer
resolved_vpa	String	-	Response: Resolved VPA (if identifier_type = MOBILE)
account_holder_name	String	-	Response: Name for customer confirmation display
resolution_status	Enum	-	Response: FOUND NOT_FOUND MULTIPLE_MATCHES

POST /psp/v1/pin/set

Attribute	Value		
Purpose	Submit an encrypted PIN block to set or reset the customer's UPI PIN		
Owning System	PSP Switch		
Target Timeout	≤ 3 s		
Idempotent	Yes		
Field	Type	Mandatory	Description
upi_profile_id	String	M	Request: Internal UPI profile identifier
encrypted_pin_block	String	M	Request: On-device encrypted PIN payload (never plaintext)
validation_method	Enum	M	Request: DEBIT_CARD AADHAAR_OTP
set_status	Enum	-	Response: SUCCESS FAILED

POST /psp/v1/pin/verify

Attribute	Value		
Purpose	Verify the customer's existing PIN for change-PIN or step-up scenarios		
Owning System	PSP Switch		
Target Timeout	≤ 2 s		
Idempotent	Yes		
Field	Type	Mandatory	Description
upi_profile_id	String	M	Request: Internal UPI profile identifier
encrypted_pin_block	String	M	Request: On-device encrypted PIN payload
verified	Boolean	-	Response: Verification result
failed_attempt_count	Integer	-	Response: Updated running count (BR-PIN-03)

POST /psp/v1/transaction/submit

Attribute	Value		
Purpose	Submit a payment transaction for NPCI routing after internal balance/fraud checks pass		
Owning System	PSP Switch		
Target Timeout	≤ 30 s (NPCI SLA ceiling); target ≤ 8 s		
Idempotent	No — use transaction_ref for status check, not resubmission		
Field	Type	Mandatory	Description
transaction_ref	String	M	Request: Unique transaction reference (idempotency key)
payer_vpa	String	M	Request: Payer's VPA
payee_identifier	Object	M	Request: Resolved payee VPA/account+IFSC

Attribute	Value		
amount	Decimal	M	Request: Transaction amount
txn_type	Enum	M	Request: P2P P2M SELF COLLECT_APPROVAL
encrypted_pin_block	String	M	Request: Authorising PIN block
npci_rrn	String	-	Response: NPCI-assigned Retrieval Reference Number
txn_status	Enum	-	Response: SUCCESS FAILED PENDING
failure_reason	String	-	Response: Present when txn_status = FAILED

GET /psp/v1/transaction/{transactionRef}/status

Attribute	Value		
Purpose	Poll for the terminal status of a Pending transaction (used by the Deferred Status Check job, BR-TXN-08)		
Owning System	PSP Switch		
Target Timeout	≤ 2 s		
Idempotent	Yes		
Field	Type	Mandatory	Description
transactionRef	String (path)	M	Request: Transaction reference to query
txn_status	Enum	-	Response: SUCCESS FAILED PENDING REVERSED
npci_rrn	String	-	Response: NPCI Retrieval Reference Number, if assigned
reconciliation_action	String	-	Response: Action taken, e.g., AUTO_REVERSED

9. NPCI UPI Network

NPCI-facing messages are exchanged exclusively through the PSP Switch adapter layer and follow NPCI's mandated UPI API specification (XML/ISO 20022-derived), not raw internal REST/JSON. The table below documents the business-level message types relevant to this platform at a field level; the Integration Team maintains the full NPCI wire-protocol mapping separately.

9.1 Key NPCI Message Types

Message Type	Business Purpose	Key Fields (Business Level)
ReqPay	Initiate a payment transaction (debit request)	Payer VPA, Payee VPA, Amount, Encrypted Credential Block, Transaction Reference
RespPay	NPCI response to a payment request	Response Code, RRN, Approval/Decline Reason
ReqValAdd	Validate/resolve a VPA or mobile-linked VPA	Identifier Type, Identifier Value
ReqMandate / ReqComplaint (Collect)	Create a collect request (payment request to a payer)	Payer VPA, Requester VPA, Amount, Expiry
ChkTxn	Query the status of a previously submitted transaction	Original Transaction Reference or RRN
ReqComplaint (Dispute)	Submit a dispute/complaint to NPCI's dispute management system	Original RRN, Complaint Category, Description
RespComplaint	NPCI response with complaint reference and status	NPCI Complaint Reference Number, Status

9.2 NPCI Response Code Handling

NPCI returns standardised response codes (e.g., 00 for success, and various U-, Z-series codes for specific decline reasons) which the PSP Switch maps to internal txn_status values and, where a direct mapping exists, to the customer-facing error catalogue in FRS Section 13. Codes without a direct mapping surface the generic fallback message defined in FRS Section 13.1.

10. Fraud Monitoring System

Provides real-time risk scoring in the transaction path. Must respond within 1 second to remain within overall transaction SLA (FRS API-REQ-08).

POST /fraud/v1/score

Attribute	Value		
Purpose	Request a real-time risk decision for a pending transaction		
Owning System	Fraud Monitoring System		
Target Timeout	≤ 1 s		
Idempotent	Yes		
Field	Type	Mandatory	Description
transaction_ref	String	M	Request: Unique transaction reference
customer_id	String	M	Request: Internal customer identifier
amount	Decimal	M	Request: Transaction amount
payee_vpa	String	M	Request: Resolved payee VPA
device_context	Object	M	Request: Device ID, IP/network signals, behavioural metadata
risk_score	Integer (0-100)	-	Response: Computed risk score
decision	Enum	-	Response: ALLOW STEP_UP DECLINE
rule_triggers	Array	-	Response: List of triggered rule codes for audit

On a Fraud Monitoring System timeout or unavailability, the PSP Switch applies the Risk Team's configured fallback policy (default-deny with manual review queue), per FRS Section 12.2.

11. AML System

Screens transactions exceeding Bank-defined AML policy thresholds, either synchronously in the transaction path or via near-real-time asynchronous screening depending on threshold and category (FRS BR-CMP-03).

POST /aml/v1/screen

Attribute	Value		
Purpose	Screen a transaction against AML policy thresholds and watchlists		
Owning System	AML System		
Target Timeout	≤ 2 s (synchronous path)		
Idempotent	Yes		
Field	Type	Mandatory	Description
transaction_ref	String	M	Request: Unique transaction reference
customer_id	String	M	Request: Internal customer identifier
amount	Decimal	M	Request: Transaction amount
counterparty_vpa	String	M	Request: Payee/payer VPA as applicable
threshold_triggered	Boolean	-	Response: Whether a screening threshold was met
screening_result	Enum	-	Response: CLEAR FLAGGED_FOR_REVIEW BLOCKED
screening_ref	String	-	Response: Reference for manual review tracking, if flagged

12. Audit Logging

Captures an immutable event record for every transaction, registration, PIN, dispute, and administrative action, retained for a minimum of 10 years (FRS NFR-CMP-04).

POST /audit/v1/events

Attribute	Value		
Purpose	Write an immutable audit event record		
Owning System	Audit Logging		
Target Timeout	≤ 1 s (asynchronous, fire-and-forget with delivery confirmation)		
Idempotent	Yes		
Field	Type	Mandatory	Description
entity_type	String	M	Request: e.g., TRANSACTION, DISPUTE, UPI_PROFILE, UPI_PIN
entity_id	String	M	Request: Identifier of the affected entity
event_type	String	M	Request: e.g., CREATED, STATUS_CHANGED, PIN_LOCKED
actor	String	M	Request: Customer ID, internal user ID, or SYSTEM
event_payload	Object	M	Request: Event-specific detail (never includes PIN or full card numbers)
audit_id	String	-	Response: Generated audit record identifier

GET /audit/v1/events

Attribute	Value		
Purpose	Query audit events for a given entity, date range, or actor (internal/Compliance/Audit use only)		
Owning System	Audit Logging		
Target Timeout	≤ 3 s		
Idempotent	Yes		
Field	Type	Mandatory	Description
entity_type	String (query)	O	Request: Filter by entity type
entity_id	String (query)	O	Request: Filter by specific entity
date_from / date_to	Date (query)	O	Request: Date range filter
events[]	Array	-	Response: Matching audit event records

13. Sample Payloads (Illustrative)

The following abbreviated JSON examples illustrate two representative calls. Full schemas with all optional fields and validation constraints are maintained in the Integration Team's OpenAPI specification.

13.1 PSP Switch — Submit Transaction (Request)

Field	Example Value
transaction_ref	"TXN-20260709-000123456"
payer_vpa	"aniket.k@dbsindia"
payee_identifier.type	"VPA"
payee_identifier.value	"merchant.store@icici"
amount	1500.00
txn_type	"P2M"
encrypted_pin_block	"<base64-encoded encrypted block>"

13.2 PSP Switch — Submit Transaction (Response, Success)

Field	Example Value
status	"SUCCESS"
npci_rrn	"312345678901"
txn_status	"SUCCESS"
correlation_id	"c7e1f9a2-..."

13.3 CBS — Debit Account (Request)

Field	Example Value
amount	1500.00
transaction_ref	"TXN-20260709-000123456"
narration	"UPI payment to merchant.store@icici"

13.4 Error Response Example (Insufficient Balance)

Field	Example Value
status	"FAILURE"
error_code	"ERR-TXN-007"
error_message	"Insufficient balance in the selected account."
correlation_id	"c7e1f9a2-..."